

PATENT SPECIFICATION

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Application Date: Oct. 23, 1939. No. 28480/39.

535,860

Complete Specification Accepted: April 24, 1941.

COMPLETE SPECIFICATION

PATENTS ACT, 1949

SPECIFICATION NO. 535,860

5 In accordance with the Decision of the Superintending Examiner, acting for the Comptroller-General, dated the twentysecond day of July, 1954, this Specification has been amended under Section 29 in the following manner:-

- 10 Page 1, line 79, after "to the" insert "reciprocal of the".
 Page 2, line 84, after "5" insert "and".
 Page 3, line 45, for " $a\sqrt{2}$ " read " $a/\sqrt{2}$ ".
 line 66, delete " $a\sqrt{2}$ " insert "the noise".
 lines 79-81, delete "This improvement width".
 15 lines 114-115, delete "noise of the receiver can alone".
 insert "received noise can only"
 Page 5, line 56, delete "by" insert "from".
 line 57, delete "(or n_1)".
 20 Page 6, line 4, delete "oscillator B" insert "multivibrator A".
 line 5, delete "100" insert "1/100".
 line 6, delete "A" insert "B".
 Page 11, line 41, delete "a substantially exact"
 25 insert "an approximate".

Attention is also directed to the following Printers Errors:-

- 30 Page 4, line 22, for "signal" read "single".
 Page 5, line 58, for " λ_2 " read " λ_1 ".
 Page 8, line 31, for " $N = E_2^{p-1}$ " read " $N = \Sigma 2^{p-1}$ ".
 line 101, for " $N = E_2^{p-1}$ " read " $N = \Sigma^{p-1}$ ".
 Page 11, line 82, for "valves" read "values".

35 THE PATENT OFFICE,
 19th September, 1955

DB 81409/1(5)/3419 100 9/55 R

40 peculiar to the transmitter, for example, ordinary telephone systems.

The information required must normally be transmitted in systems where interference noises of an amplitude which varies with the system are inevitable. In systems of the type (a) means are already known for ensuring that this noise should always remain below a certain predetermined maximum value, and have practically no effect on the results received. The solution generally adopted consists in the use of a relay, or other device with trigger operation.

The wave form to be transmitted is divided into a finite number of predetermined amplitude values according to the degree 95 of fidelity required, the instantaneous amplitude value of the wave form to be transmitted at each predetermined instant being transmitted in a signal code representing the nearest predetermined amplitude value above or below said instantaneous amplitude value. 100

The number of elements in the code may equal the number of predetermined values of the amplitude range of the 105 complex wave, or may be equal to a

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Electric Signalling Systems

(Communicated by LE MATERIEL TELEPHONIQUE, SOCIETE ANONYME, a French Company, of 46, Quai de Boulogne, Boulogne-Billancourt, Seine, France, and of 46, avenue de Breteuil, Paris, VII^e, France.)

We, STANDARD TELEPHONES AND CABLES LIMITED, a British Company, of Connaught House, 63, Aldwych, London, W.C.2, England, do hereby declare the nature of this invention and in what manner the same is to be performed, to be particularly described and ascertained in and by the following statement:—

The present invention relates to electrical signalling systems, and more particularly to systems adapted to transmit complex wave forms, for example, speech, which are practically free from any background noise.

The main object of the invention is to provide electrical signalling systems which practically have no background noise, even under conditions in which the signal-to-noise ratio would normally be about 20 decibels, or less.

The systems for the transmission of intelligence at present in use may be grouped into two large classes: (a) systems giving at the receiver indications of the occurrence or non-occurrence of actions taking place at the transmitter, for example, of the presence or absence at any given moment of particular pulses forming a part of the signals transmitted by a teleprinter transmitter, and (b) systems giving the receiver a quantitative information with regard to the amplitude at the moment in question of a variable peculiar to the transmitter, for example, ordinary telephone systems.

The information required must normally be transmitted in systems where interference noises of an amplitude which varies with the system are inevitable. In systems of the type (a) means are already known for ensuring that this noise should always remain below a certain predetermined maximum value, and have practically no effect on the results received. The solution generally adopted consists in the use of a relay, or other device with trigger operation,

adjusted in such a way that although actuated by the desired signal the noise, always of less amplitude than the signal, can never cause operation of said relay, or prevent its operation under the effect of the desired signals.

In the case of systems of type (b), however, the methods employed to overcome the noise effects are only palliatives for reducing the noise, but do not eliminate the noise entirely.

It is, consequently, the main object of the invention to provide means to eliminate the background noise of a signal transmission substantially completely in systems of the second of the two above-mentioned classes, and to provide arrangements for this purpose which have the desired commercial characteristics. The arrangements proposed necessitate an increase of the width of the transmission frequency band.

This necessary increase of the band width is in proportion on the one hand to the signal-noise ratio before the elimination of the noise, and on the other hand to the precision necessary for the reproduction at the receiver of the signal wave form at the transmitter.

According to the present invention, a signalling system for transmitting complex wave forms, for example speech, wherein the wave form is scanned at the transmitter at predetermined instants, and at these instants signal elements dependent on the wave form are transmitted to the receiver is characterised in this, that the amplitude range of the wave form to be transmitted is divided into a finite number of predetermined amplitude values according to the degree of fidelity required, the instantaneous amplitude value of the wave form to be transmitted at each predetermined instant being transmitted in a signal code representing the nearest predetermined amplitude value above or below said instantaneous amplitude value.

The number of elements in the code may equal the number of predetermined values of the amplitude range of the complex wave, or may be equal to a

number such that the number of combinations of the code elements is equal to or greater than the finite number required for reproducing the complex wave form within the desired limits of accuracy.

In accordance with another feature of the invention, the amplitude of the complex wave form is transmitted at a given moment by a predetermined combination of m separate signalling channels, each having n different signal code elements, the number of code element combinations, that is to say n^m being equal to or greater than the finite number required for the transmission of the information within the desired limits of precision.

In accordance with another feature of the invention, the combinations of code elements are automatically transformed at the receiver into an approximate reproduction depending upon the fidelity required of the instantaneous amplitude at the moment in question of the wave form to be communicated.

In accordance with another feature of the invention, the number of possible code elements transmitted in each channel is reduced to a value at which the circuit noise in each of these channels of, a peak value not exceeding a predetermined limit, is incapable of affecting the occurrence or non-occurrence in the receiver of correct and definite series of operations associated with the particular combination of code elements transmitted at that moment.

In accordance with another feature of the invention, the m separate signalling channels are obtained by employing mn channels each utilising a different carrier frequency.

In accordance with another feature of the invention, the m signalling channels are obtained by employing a carrier frequency common to all the channels, and utilising a distributor system for the channels in known manner.

In accordance with another feature of the invention the n different code elements on each channel are obtained by the transmission of the carrier frequency of each channel at n different predetermined amplitudes.

In accordance with another feature of the invention, the n different code elements are obtained by the transmission on each channel of n pulses of different duration or n pairs of pulses having n predetermined different intervals of time between the pairs.

In accordance with still another feature of the invention, the n different code elements are obtained on each channel by the transmission of the carrier

wave of each channel at n possible different phases.

In accordance with still another feature of the invention, the n different code elements are obtained by the use on each channel of n different predetermined frequencies.

In one practical embodiment of carrying out the invention, in the case in which the number of channels m is equal to 1, a multi-vibrator associated with an oscillation generator synchronised with an oscillation generator at the transmitter and associated with a frequency band filter is employed at the receiver for reproducing the signal elements.

In a practical embodiment of the invention in the case in which m is equal to 5 n is equal to 2, a frequency divider circuit associated with suitable triggering devices is used at the transmitter for producing codes representing amplitudes of the waves and at the receiver for re-converting the codes into amplitudes.

These, and still other features will be explained in detail in the following description based on the attached drawings in which:

Fig. 1 shows an example of the complex wave form of a signal to be transmitted;

Fig. 2 shows an example of a transmitting circuit incorporating features of the invention;

Fig. 2A shows curves used in the explanation of the operation of the circuit shown in Fig. 2.

Fig. 3 represents an example of a receiving circuit embodying certain features of the invention;

Figs. 4 and 5 are circuit diagrams of another example of transmitting and receiving circuits respectively incorporating features of the invention.

In a telephone transmission system, a wave form such as that of Fig. 1 can, as is wellknown, (for example, as in transmission by means of individual pulses) be transmitted or received with negligible distortion by the successive transmission and reception of a number of separate pulses, each corresponding in amplitude (or in any other desired variable factor) to the original instantaneous amplitude of the speech, as indicated by the small circles on the curve of Fig. 1, the spacing between the pulses being considerably less than that corresponding to a period of the highest frequency contained in the speech to be transmitted.

In each possible case arising in practice, however, each of these amplitudes has to be transmitted, not with an infinite degree of precision, but with a particular finite degree of precision de-

pendent upon the problem involved. For example, in considering ordinary speech signals at a mean constant level, a precision of the order of $\pm 1.5\%$ should,

5 in the ordinary way, be amply sufficient in practice, while a precision of $\pm 0.05\%$ would certainly be sufficient for a high quality or radio broadcast of music. Consequently, taking the case of 1.5% in
10 which the percentage is relative to the maximum amplitude it is clear that in this case it is unnecessary to transmit an infinite possible number of amplitudes, but only a series of approximately
15 33 different values. Consequently, if there is used at the transmitter a marginal relay—one for each of the 33 possible different amplitudes mentioned above, and such that only the amplitudes
20 of signals falling in the range of $\pm 1.5\%$ of its allotted value cause it to operate, only 33 finite different amplitudes alone need be transmitted and if, moreover, 33 similar relays are provided at the receiver
25 each will only respond to a single pulse of the transmitting relay, the original wave form will be received with the desired precision.

The background noise effect will now
30 be considered. Assuming that the band-width of the audio frequency is 3000 periods per second, and that the normal signal-to-noise ratio when a linear receiver is employed with a minimum
35 band-width is a (ratio of the potentials), whatever the frequency of the modulating wave, the minimum band required to transmit the resulting amplitudes at the series of instants (only taking a single
40 side-band if the said amplitudes are employed to modulate a high frequency carrier wave) will be $2 \times 3000 = 6000$ cycles. The signal-to-noise ratio for the amplitudes received will, consequently,
45 be $a\sqrt{2}$.

In order completely to eliminate the noise in the receiver output, the noise peaks must always be lower than the value necessary to change the amplitude
50 of the received pulse from its actual value to a value of a higher or lower degree in the series; that is to say, the noise peaks must be less than 1.5% of the maximum pulse amplitude,—a value of 37 decibels
55 below the pulse potential. The effective level (root-mean-square) of the noise corresponding to this peak value of course depends upon the wave form of the noise; if, however, this noise is not of the nature
60 of a steady hiss caused by thermic agitation which is the limiting noise factor in a receiver of ultra-short waves correctly designed under normal conditions, a peak factor of $\sqrt{2} = 3$ decibels may be
65 assumed. For practically complete

noise suppression, consequently, $a\sqrt{2}$ has a maximum effective value of 40 decibels below the maximum pulse potential.

Assuming a safety factor of three additional decibels it is found that it is
70 unnecessary for the original normal signal-to-noise ratio on an ideal linear receiver to exceed 46 decibels. If, however, this condition is fulfilled the output of the receiver will give a signal-to-
75 noise ratio of 60 decibels or more, the limit only being determined by the noise generated by the receiver marginal devices themselves. This improvement has been made by doubling the band-
80 width.

Another case can now be considered in which the band-width may be increased indefinitely (within reasonable limits, for
85 example, in a ratio of 30 if necessary) so as to obtain the best possible signal-to-noise ratio from the minimum possible incident field strength. Such a case is frequent in communication by ultra-
90 short waves. It will be assumed that 32 amplitude values are necessary, i.e. approximately about as many as above. It is possible to obtain the necessary permutations of code elements in order to
95 represent 32 different amplitudes by suitable combinations of five separate pulses used as the code elements, each having only two possible values—either “on” or “off”—and the circuit being
100 designed in such a way that the correct combination of pulses is automatically transmitted, received and automatically re-translated at the receiver into the 32 original amplitudes, as will be explained
105 hereinafter. Each pulse requires a separate transmission path. This can be carried out in a large number of different ways by employing frequencies individual to each channel, by employing the principle of distribution in time etc.
110 The five corresponding trigger devices of the receiver now consist of simple relay devices each having an “on” and an “off” position; and the noise of the receiver can alone prevent perfect opera-
115 tion (if the adjustments are optimum), if the noise peaks exceed half the amplitude of the pulses. While assuming, as hereinabove, a safety factor also of 3 decibels, a substantially perfect recep-
120 tion will be obtained on each of the pulse channels if the effective signal-to-noise ratio (root-mean-square) on these channels is equal to, or greater than, 12 decibels. The energy of the transmitter
125 is divided between five channels and each channel, in consequence of the pulse system employed, normally requires, as mentioned above, twice the band-width
130 necessary for speech alone. In order to

give 12 decibels on each of the five pulse channels it is necessary to have an original normal signal-to-noise ratio of $12+10=22$ decibels. As the output of the receiver, however, may now have a signal-to-noise ratio of 60 decibels or more an improvement of at least 38 decibels has been obtained—at the expense of an increase in the band-width by a factor of 10 only.

Another improvement can be obtained by utilising either of the known pulse modulation systems, either single pulse modulation or double pulse modulation systems.

In the pulse modulation system the signals are transmitted as pulses of constant amplitude, the duration of a pulse being proportional to the instantaneous amplitude of the signal wave form represented by the pulses. This system is known as the signal pulse modulation system and in a modification an extremely short pulse is transmitted to mark the beginning and end of each of the pulses of variable duration. When the beginning of the variable duration pulses takes place at equal intervals of time, or when the end of the variable duration pulse takes place at equal intervals of time, either the marking pulse, that is the pulse of short duration marking the beginning of a variable duration pulse, or the trailing pulse, that is the pulse of short duration marking the end of a variable duration pulse, whichever is equally spaced in time, may be eliminated at the transmitter and reinserted again at the receiver. These modified single pulse systems are known as double pulse modulation systems. Whatever the pulse modulation system employed, if the band-width used for each pulse channel can be increased in a ratio equal to n , it is possible to employ a pulse wave-form giving to the useful portion of the pulse a duration which is only $1/n$ of the total time allocated to its transmission. Although the peak energy of the pulse is then the same as before, its mean energy is only $1/n$ of its original value. If, consequently, a type of valve is employed in which the limit is the mean heat dissipated rather than the peak power, within certain limits, as it is ordinarily possible to do in a typical ultra-short wave transmitter, then it is possible to increase the peak power by multiplying it by the factor n without overloading the valve involved. In a practical case if $n=5$, for example, by this method, a quantity equal to the original total peak energy will be available for each of the five channels. The band-width required for each channel is

increased in the ratio 5, thus increasing the total noise energy in the receiver also in this ratio. During $4/5$ of each period available for the pulses, however, the receiver is entirely inoperative and the noise has no effect. Consequently, the total noise energy which tends to disturb the action of the circuit of the trigger devices of the receiver will not be affected by the increase of the band width, giving a signal-to-noise ratio equal to that which would be obtained if the energy were concentrated on one channel only. Thus, a normal signal-to-noise ratio is required, measured under normal conditions with an ideal receiver, of 15 decibels instead of 22 decibels as before, the subsequent improvement being obtained by an additional increase of the band-width in a ratio of 5, a condition of ultra-short wave operation which can frequently be obtained in practice without inconvenience.

Neglecting now the improvement above due to the use of valves in which the output is dependent upon the mean heat dissipated rather than the peak energy, in the first place if 32 possible amplitudes are not sufficient for any particular case, but the suppression of the noise is still required with an initial minimum signal-to-noise ratio, then the number of pulse channels must be increased to a number m , in which 2^m is equal to the number of combinations required, i.e. one combination of pulses for each amplitude position required (m being taken equal to the nearest integer above the exact value given by the formula). For example, for 1,000 positions, m will be equal to 10, which gives $2^m=1,024$. This requires a factor of increase of band-width of 20 instead of 10, giving an initial signal-to-noise ratio of 25 decibels instead of 22. If, moreover, 32 positions are still sufficient, but an initial signal-to-noise ratio of 30 decibels is available instead of 22 decibels, it is clearly inefficient to employ the 5 pulse channels mentioned above. This case would be a compromise between the example with five channels given above and the first case dealt with (that of a single pulse channel and an initial ratio of 46 decibels). As there is now an excess of 8 decibels of signal-to-noise ratio available, assuming five pulse channels, it is possible to employ in each pulse channel a device responsive to four values of amplitude instead of two only "on" and "off", and still to have two decibels in reserve. m should then be chosen so that $4^m=32$. The nearest integer above the exact value is $m=3$ and $4^m=64$. Consequently, three

channels must be used, each transmitting pulses of four amplitude values, and thus give a safety factor of $\sqrt{5/3}$ due to the reduced noise per channel.

- 5 In considering now a formula for the general problem—as a first step concerning the case of the smallest possible initial signal-to-noise ratio, in which each pulse should have two amplitude values only “on” and “off”, the minimum signal-to-noise ratio for each separate impulse channel is 2θ (ratio of volts) in which θ is the safety factor adopted (for example, 1.4). Assuming that there are 2^m necessary amplitude positions (total) in which m as before represents the nearest integer above the exact figure, then m pulse channels will be necessary, and the initial signal-to-noise ratio on an ideal linear receiver will be $2\theta \times \sqrt{\text{factor of increase of the band-width}}$, that is to say; initial ratio $= a = 2\theta \sqrt{2^n m}$.

- Considering now the general case in which there is an initial given signal-to-noise ratio of a , and a required number of amplitude positions equal to N , in order to find the optimum or other number of amplitude positions or alternatively the number for each pulse channel, and the total factor of increase of band-width giving a complete suppression of noise, then if each channel transmits pulses of n different amplitudes, and there are m pulse channels as before, the total number of pulses representing amplitude positions will be equal to n^m ; this figure is equal to N .

- The problem can be more conveniently dealt with by successive approximations. Let in the first instance $n=2$. It is then found that the number of pulse channels $m_1 = \log_2 N$, and the band-width must consequently be increased by the factor $2 \log_2 N$. (m_1 being as above, the nearest integer above the exact value). The signal-to-noise ratio per pulse channel will then be

$$a / \sqrt{2^n m_1}$$

- 50 The minimum signal-to-noise ratio required per channel is 2θ ; the margin of the signal-to-noise ratio is consequently equal to the factor

$$\frac{a}{2\theta \sqrt{2^n m_1}} = \lambda_1$$

- 55 If this latter factor is not less than 1.5 it is then possible to increase n by the value 2 (or n_1) to a value n_2 , such that $n_2/2$ is the nearest possible integer to λ_1 , while n_2 remains an

integer.

In this way, for m a new integer m_2

is obtained such as $n_2^{m_2} = N$, that is to say, the new value (m_2 being again the nearest integer higher than the exact value) $m_2 = \log_{n_2} N$. The signal-to-noise ratio available per channel will now be greater than before; it will have the

$$\text{value } \frac{a}{\sqrt{2^n m_2}}$$

The new marginal factor λ_2 will then

be equal to $\frac{a}{2\theta \sqrt{2^n m_2}}$. If $2\lambda_2$ is now

greater than $(n_2 + 1)$, the above operation will be repeated p times until the final figure $2\lambda_p$ is less than $(n_p + 1)$; n_p is then the optimum number of positions required per pulse channel, m_p is the optimum number of pulse channels; and the factor of increase of band-width is equal to $2m_p$.

Certain details of the circuits employed for utilising the invention will now be described.

First of all, the arrangements according to the invention intended to eliminate noise in a circuit having signal-to-noise ratio a of 48 decibels, and in which 32 amplitude positions are sufficient, will be described.

As will be seen, in accordance with the general formula, and the case hereinbefore described, the most efficient solution in this case is to employ a single pulse modulation system having 32 different code elements for the same number of amplitude positions, which again will give an additional signal-to-noise margin of 2 decibels.

A suitable form of transmitter circuit for this case is shown in Fig. 2. In this circuit the potential of the incoming line is applied as shown in push-pull relation to the two grids of the double triode A which consists of a valve adapted to oscillate as a multi-vibrator at a frequency outside the speech frequency range employed (for example, at 6000 cycles) by means of its associated circuit. The effect of speech is to modulate in duration almost linearly, the resultant rectangular plate current pulses, that is, the ratio of the portion “on” to the portion “off” in each period of plate current is proportional to the potential of speech at the moment concerned, while the amplitude of these rectangular pulses remains constant.

The auxiliary oscillator B is loosely coupled by the variable condenser C to one of the plates of the double triode,

and the frequency of this oscillator B is adjusted so as to be approximately 100 times that of A. As is well known, the oscillator B can be caused in this manner to lock itself in operation at 100 times exactly of the frequency of A (or at any other integral ratio provided that the wave form of A is sufficiently steep, which condition usually arises without the provision of special arrangements).

As the frequency ratio is 100/1 there will be more than 100 different stable phase relations between the two oscillators.

The coupling of B to A is arranged to be such that each signal of the multivibrator A can only begin at the moment of reception of a peak from B.

While the input current at speech frequency causes the moment of inception of these multivibrator pulses to vary, there will now only be a definite number of possible inception times corresponding to the variable phase relations between the multivibrator and the oscillator B; and if the maximum of time modulation by speech is 64% (normal and practical amount) there will be 32 possible values of percentage of modulation in time, as the value of the percentage varies from the maximum to zero. For any level of the incoming signals at speech frequency other than the exact level normally giving one of the 32 stable time modulation values, there will result a time modulation which is the next lower stable value nearest the exact value corresponding to the momentary speech amplitude. By combining the present invention with the system of pulse modulation, the modulation in definite steps desired has been obtained.

The output from the two plates of the multivibrator is then applied through the two small coupling condensers D and E both shunted to earth through the metal rectifier F. The operation of this circuit is given below. In this way, for each period of the multivibrator the time interval between these pulses only has 32 possible values, and is proportional to the speech amplitude at the moment concerned.

The operation of Fig. 2 is as follows:—
Let curves (a) and (b), Fig. 2A, represent the current wave-forms on the left and right-hand plates respectively of the multivibrator A (Fig. 2).

It will be noticed that in each case the rise of current is more rapid than its fall. This is due to the fact that with usual constants a multivibrator gives a high positive kick on to the opposite grid at one point of the cycle of events; and this high positive grid voltage causes a

momentary value of plate internal impedance low compared with the normal load resistance (say 50,000 ohms for the latter). The time-constant of the rise of plate current is determined by the rate of charge of the capacity of the load (self or stray), and is substantially independent of the load resistance—the rate of charge through the internal (low) tube impedance). The time-constant of the fall of plate current, however, is determined by the rate of discharge of this same load capacity through the load resistance,—a higher resistance value.

Original "double-pulse" circuits used two rectifiers and two resistances, but the above considerable difference between the times of rise and fall of the plate current has enabled in some cases, the simpler circuit of this Fig. 2 to be used.

(c) shows the derived curve of (a)—it is the voltage across the output resistance (R) given by the left-hand plate alone in the absence of the rectifier:

(d) shows the corresponding curve for the right-hand plate again in the absence of the rectifier. It will be noticed that because of the unequal rise and fall times of the plate currents the positive kicks in (c) and (d) are in every case smaller in amplitude than the negative kicks, and also displaced slightly in time from the latter. For this reason if these two derived curves are added together—as occurs across the common output resistance R—instead of getting zero (as is the case if the rise and fall times were exactly equal), the curve (e) is obtained. The rectifier then eliminates the resulting small positive kicks, and the correct negative double pulse across R, as shown in curve (f) is obtained.

The output from the multivibrator is then applied to a high frequency modulator (not shown) in known manner and at the same time a small fraction of the potential at 600 kc. (or a sub-harmonic) for synchronising purposes at the receiver is transmitted as a pilot frequency.

Fig. 3 shows an embodiment of a receiver adapted to operate in conjunction with the transmitter illustrated in Fig. 2. The double triode AR is incorporated in a well-known multivibrator circuit employed in association with the low-pass filter BR and the coupling valve CR in order to produce a normal speech-wave form from the pulses arriving at DR after amplification from the "linear" portion of the receiver. The auxiliary oscillator ER is synchronised with the auxiliary oscillator B of the transmitter (Fig. 2) by employing the said pilot frequency sent from the transmitter and which is selected and amplified at the

receiver. If a sub-harmonic of the 600 kc. pilot frequency from the transmitter be employed instead of the frequency of the latter itself, synchronisation is accomplished by using a suitable harmonic generator apparatus in the synchronising circuit of the receiver, or rather if desired, the oscillator ER may be eliminated, the pilot frequency being amplified and selected, or one of its harmonics, and directly employed.

The oscillator ER, or its equivalent is loosely coupled to the input circuit of the multivibrator as shown, by the condenser FR, the degree of coupling and the other constants involved being chosen in such a way that although neither the received pulses, nor the oscillator ER can separately actuate the multivibrator, the actuation is accomplished when these two potentials act in agreement.

The two 600 kc. oscillators have such a fixed phase relation that a pulse transmitted at the moment of a peak of the oscillation of B at the transmitter arrives at the receiver (taking into consideration the time of propagation)—also at a peak of the oscillation of ER at the receiver. As the circuit of the transmitter is established so that the output pulses are always produced at such moments, these pulses will always consequently be received exactly at the moments of sensitivity of the receiver. In systems of pulse modulation such as are employed in the present invention, the noise may again slightly affect the receiver unless the pulses are infinitely steep in wave form by slightly modifying the building-up time of the received pulses, but in the arrangement according to the invention the exact moment of the triggering operation is determined solely by the moment of arrival of the local pulses from the receiver oscillator ER oscillating at 600 kc. Pulses may be given as steep a wave-front as desired without further increasing the band-width employed in the transmission. The effect of the speech modulation is to change the triggering moment when the receiver multivibrator is actuated from one of its definite positions to the other, and this action is quite independent of the residual circuit noise which, consequently is eliminated.

Arrangements for eliminating noise in a circuit having an initial signal-to-noise ratio α of 22 db. and in which 32 amplitude positions are sufficient will now be examined;

As stated hereinabove, the correct solution in this case is to employ five pulse channels, each channel simply having an "on" and "off" condition, the

resultant factor of increase of band-width being 10.

Fig. 4 shows a form of transmitter particularly provided for this purpose. The valve AT is a double triode associated with a multivibrator circuit, giving a pulse modulated output of the speech line BT, as in Fig. 2, but without the 600 kilocycle oscillator. The output of this valve AT is associated with an arrangement for counting the number of cycles of a specific frequency which occur during the respective pulses in the output of AT. This counting operation is attained by the use of a frequency divider arrangement having a plurality of stages of the kind described fully in Specification No. 296,827, in cascade. Five stages are shown in Fig. 4 and each comprises a multivibrator circuit having respectively double triode valves C1, C2, C3, C4 and C5. The output of an oscillator DT operating at 600 kilocycles, for example, is supplied at FT through push-pull coupling valve ET to the input of the first frequency divider stage C1. A pair of grids of ET, connected to the multivibrator AT as shown, is polarised in such a way that ET is only active during the positive periods of the grid GT of the multivibrator AT. The conditions are adjusted so that during the longest positive period of GT resulting from the speech, 32 complete periods of DT will take place. If the modulation in time of the multivibrator AT is 100%, any number of periods of ET between 0 and 32 can be produced according to the speech potential at a specified moment. If the modulation in time is not 100% it is arranged that the difference between the numbers of periods of ET corresponding to maximum and minimum modulations respectively is 32, but the first and simplest case will now be described in order to facilitate the explanation of the action of the device.

Just before the beginning of a positive half cycle of GT it is arranged that all multivibrators of the five stages are in the first position No. 1 of equilibrium, that is to say, that the plate current passes in all the left-hand plate output circuits of the multivibrator valves C1—C5, but that there is no current in the right-hand plate output circuits. At the beginning of the positive half-cycle of GT and until its end, the coupling valve ET will function, to allow the oscillator DT to feed the first of the frequency divider stages, each of which reduces the frequency applied to it for example, by one half. The output frequency from the respective stages, beginning at the first is therefore $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$ and $\frac{1}{32}$

respectively of the frequency of DT.

The frequency at the output of each stage is one half the frequency at the input. Thus if C2 has changed over to its second condition of equilibrium, C1 must have completed one cycle, that is, two cycles of DT have been transmitted, that is 2^1 or 2^{2-1} . If C3 has changed over to its second condition of equilibrium, C2 must have completed one cycle, or two cycles of C1; that is 2×2 cycles of DT have been transmitted, that is 2^2 or 2^{3-1} . Similarly if C4 has changed over to its second condition of equilibrium, the number of cycles of DT which have been transmitted is $2^{4-1} = 2^3$. Generally, the number of cycles of DT which have been transmitted to change any multivibrator from its first to its second condition of equilibrium is given by the formula $n = 2^{p-1}$ in which p represents the order in succession (starting with $p=1$ for the stage C1). At the end of a counting operation, namely at the end of a positive half cycle of GT the total number of cycles of DT which have been transmitted is the sum of the cycles which have been transmitted to change those stages which are in their second condition of equilibrium at the end of the counting operation, i.e. $N = \epsilon 2^{p-1}$. For example, if the stages 2, 3 and 5 are in position No. 2, with the stages 1 and 4 in position No. 1, the number of cycles transmitted is then:

$$2^{2-1} + 2^{3-1} + 2^{5-1} = 22.$$

There is thus, obviously, a particular combination of positions of final equilibrium corresponding to any number of cycles transmitted between zero and 32, that is to say, to all possible numbers in the case in question.

The small condenser HT, with the resistance-rectifier combination JT, is arranged to give a sharp negative pulse to the right-hand grid of the double triode KT at the moment of the end of the positive half period of GT. KT is associated with a trigger circuit, actuated by negative pulses on the right-hand grid LT of KT, having a short time of operation compared with a half-period of the oscillations produced in DT, and a time of restoration (adjusted by the constants of the resistance-rectifier in condenser network MT) equal to slightly less than half the time elapsing between the successive positive half-cycles of AT when the latter have their maximum duration, (a form of such a circuit with trigger operation is shown in the drawing).

As shown in the diagram, the pulse

produced by the restoration of KT is applied to a similar trigger circuit NT having a restoration time of the same order as that of KT so that NT is always re-established before the next positive half-period on the grid of GT.

The five pulse channels automatically transmitting in code to the antenna Q, the instantaneous speech amplitude at the input originate in the five master oscillators O1, O2, O3, O4, and O5, and are amplified in respective amplifiers P1, P2, P3, P4 and P5 by application to the respective grids 3 thereof, and passed to the antenna Q. These amplifiers are so biased as only to operate when their two inner grids 1 and 2 are simultaneously made less negative. One grid 2 of each amplifier is connected in parallel with the right-hand grid of the corresponding multivibrator valve C; the grids 1 of the amplifiers are all connected in parallel through a compensation battery R to the right-hand plate of the valve of the trigger circuit NT.

Consequently, each amplifier can operate only when the corresponding multivibrator stage is in position NO.2 at the same time as the operative circuit NT is in position "on". The particular combination of adjacent frequency pulses transmitted during this latter condition will, consequently, indicate the combination of equilibrium of the multivibrator stages at the end of the counting operation, and, consequently, will indicate in accordance with the formula $N = \epsilon 2^{p-1}$, the number of periods of DT which have been transmitted during the positive half-period of GT, this number directly depending upon the momentary amplitude of the speech at the moment in question.

In order to prepare the next positive half-cycle of GT the resistance-capacity-rectifier network ST is provided, which at the moment of restoring of NT, a sharp negative pulse is applied through the de-coupling circuits T, to all the right-hand grids of the frequency divider multivibrator valves. The effect of this negative pulse is to re-establish all the multivibrator stages in position No. 1, that is to say, with each valve having current in the left-hand plate circuit, but not in the right-hand plate circuit. This condition is the same as in the original conditions of the whole circuit, which consequently is ready to receive the second positive half-cycle of GT.

In summary, the complete operation is as follows:

(a) The incoming speech modulates the duration of the rectangular current pulses in the multivibrator AT, in such

a manner (as in pulse modulation) that it gives a substantially linear relation between this duration and the instantaneous amplitude of speech.

- 5 (b) The beginning of each positive half-cycle on the grid GT actuates the coupling tube ET which, in turn, permits oscillator DT (at a considerably higher frequency than the limit of the speech
10 band), to actuate the five multivibrator stages C1, C2 etc. This action continues through the positive half-cycle of GT, then stops, the final combination of equilibrium of the five multivibrator stages
15 representing as mentioned above in the code employed, the instantaneous speech amplitude at the moment in question.

- (c) As soon as the positive half-cycle ceases on GT the trigger circuit KT
20 comes into action in the position "on" and thus re-establishes its first condition after a short predetermined time interval. During this interval the temporary interferences in the multivibrator stages have
25 time to disappear and the combination of stable equilibrium to be taken up.

- (d) The impulse produced on the restoration of KT actuates the similar circuit NT in the position "on" which
30 in turn re-establishes itself in the original position just before the next positive half-cycle of GT. While NT is in the position "on" the particular amplifier tubes P1, P2 etc. which are
35 connected to multivibrator stages in position of equilibrium No. 2, allow the master oscillator O1, O2 etc. on adjacent frequencies to transmit the required combination of code pulses through the
40 antenna Q. The other pulse channels remain unchanged.

- (e) The pulse due to the restoration of NT triggers all the multivibrator stages C1—C5 back to their position No.
45 1, thus preparing them for the second positive half-cycle of GT, the cycle of operations described above being repeated indefinitely.

- Fig. 5 shows a form of receiver circuit
50 adapted to operate in conjunction with the transmitter of Fig. 4. AL represents the first portion of a receiver of usual and known type connected to an antenna; the output contains a suitably amplified reproduction of the five pulse channels
55 from the transmitter. The channel 1 is then selected by the filter B1, the channel 2 by B2, etc. each being selected by its respective filter. An oscillator DL of approximately the same frequency,
60 that is to say 600 kilocycles, as that of DT of the transmitter Figure 4, is connected through the coupling valve EL to the input of the five-stage frequency
65 divider LC1, LC2, LC3, LC4, LC5

similar to that described in connection with Figure 4. At the beginning of the sequence of operations all the stages are in the position of equilibrium No. 2, that is to say, with currents passing in
70 all the right-hand plate circuits, and no current in the left-hand plate circuits; the coupling valve EL is blocked. As soon as the combination of pulses arrives
75 from the transmitter, those of the pulse channels entering in the combination are each applied to the corresponding stage through the appropriate filter B1—B5. Each of these filters also contains
80 a rectifier circuit; consequently, when the receiver is in operation, the outputs from the filters produce single pulses through the respective condensers F1—F5, on the left hand grids of the respective
85 multivibrator valves LC1—LC5. The pulses applied to the relevant multivibrators are arranged to be of positive sign to fulfil the conditions described in Specification No. 296,827, when the pulse
90 is applied to one grid of the stage only, and the result is that all the multivibrator stages thus affected will be triggered from position No. 2 to position No. 1, and will remain in position No. 1.

Upon the arrival of any pulse combination the trigger circuit GL, of similar design to that of KT and NT in Fig. 4, is actuated, and in turn the pulse generated on its restoration to position
95 No. 1 (after a short predetermined interval of time) operates trigger circuit HL to the position "on". This latter is of different design from that of GL and is as shown. It consists of a double
100 stability device, of a time-constant in both directions which is short in proportion to a half cycle of DL. In the position "off" HL only has current in its
105 left-hand plate circuit and in the position "on" it only has current in its right-hand plate circuit. The operative pulse from GL to HL, is consequently
110 of negative sign, when it is applied as shown to the left-hand grid of HL. When HL is "off" the inner grid of
115 EL is sufficiently negative to prevent the coupling valve EL from operating. When HL is "on", however, the increased positive potential on the left-hand plate of HL causes EL to give the
120 correct amplification for operating LC1 and thus initiates the counting operation of the frequency divider which determines the duration of HL in its "on" position by counting a predetermined
125 number of cycles of the oscillator DL. This counting operation is as follows:—

Taking the example described with reference to Fig. 4; stages 2, 3 and 5 will be changed over to their first positions.
130

Stage LC1 is therefore in its second position with current in its right hand anode circuit. It will start to operate on the first negative half wave from DL.

5 When it changes over from position No. 2 to position No. 1 a positive pulse is applied to the grids of LC2, but LC2 is not affected by the positive pulse and remains in its first position. On the

10 reception by LC1 of the next negative half wave of DL, LC1 changes over from its first position to its second position and generates a negative pulse on its right hand anode, which is applied to LC2 and

15 causes this latter to change over from position No. 1 to position No. 2 generating a negative pulse on its right hand anode which is applied via the coupling condensers to the grids of LC3 which is

20 changed over from position No. 1 to position No. 2 generating a negative pulse which is applied to the grids of LC4. Thus, so far one cycle of DL has elapsed. Stage LC4 changes to its position

25 No. 1 generating a positive pulse on its right hand anode which is ineffective as regards LC5. On reception of the next negative pulse from LC3, i.e. after

2³=8 cycles of DL have elapsed LC4 changes over to its position No. 2 generat- 30 ing a negative pulse on its right hand anode which is applied to the grids of LC5 and causes this latter stage to change over to its position No. 2 and generate a negative pulse on its right 35 hand plate, and a positive pulse on its left hand plate. This positive pulse is applied as described more fully herein- after to terminate the counting operation to the left hand grid of the double 40 stability circuit HL and triggers it over to its position No. 1, thus increasing the negative bias on the inner grids of EL, cutting off the transmission from DL to LC1 and interrupting the counting 45 operation.

The period elapsing between the instigation of the counting operation and the operation of LC5 to position 2, as will be seen from the above description of 50 the operation of Fig. 5, is 9 cycles of the oscillations produced by DL. Corresponding numbers of cycles counted at the transmitter and receiver are given in the following table:—

55

Cycles of DT counted at Transmitter (Fig. 4).	Channels transmitted No. and Stages changed over at Receiver No.	Cycles of DL counted at receiver (Fig. 5)
1 = 2 ¹⁻¹	1	30
2 = 2 ²⁻¹	2	29
4 = 2 ³⁻¹	3	27
6 = 2 ³⁻¹ + 2 ²⁻¹	2 & 3	25
8 = 2 ⁴⁻¹	4	23
10 = 2 ⁴⁻¹ + 2 ²⁻¹	4 & 2	21
12 = 2 ⁴⁻¹ + 2 ³⁻¹	4 & 3	19
14 = 2 ⁴⁻¹ + 2 ³⁻¹ + 2 ²⁻¹	4 & 3 & 2	17
16 = 2 ⁵⁻¹	5	15
18 = 2 ⁵⁻¹ + 2 ²⁻¹	2 & 5	13
20 = 2 ⁵⁻¹ + 2 ³⁻¹	3 & 5	11
22 = 2 ⁵⁻¹ + 2 ³⁻¹ + 2 ²⁻¹	2 & 3 & 5	9
24 = 2 ⁵⁻¹ + 2 ⁴⁻¹	4 & 5	7
26 = 2 ⁵⁻¹ + 2 ⁴⁻¹ + 2 ²⁻¹	2 & 4 & 5	5
28 = 2 ⁵⁻¹ + 2 ⁴⁻¹ + 2 ³⁻¹	3 & 4 & 5	3
29 = 2 ⁵⁻¹ + 2 ⁴⁻¹ + 2 ³⁻¹ + 2 ¹⁻¹	1 & 3 & 4 & 5	2
30 = 2 ⁵⁻¹ + 2 ⁴⁻¹ + 2 ³⁻¹ + 2 ²⁻¹	2 & 3 & 4 & 5	1
31 = 2 ⁵⁻¹ + 2 ⁴⁻¹ + 2 ³⁻¹ + 2 ²⁻¹ + 2 ¹⁻¹	1 & 2 & 3 & 4 & 5.	0

The pulse generated by HL has a period which is equal to the number of cycles of DL which have been counted

60 plus the "hangover" of HL. It will be observed from the above table that the variations of the periods of the pulses generated by HL, Fig. 5, are the inverse substantially of the variations of the

65 periods of the pulses in the output AT.

Fig. 4, which represent the instantaneous amplitude of the speech wave at the input of BT.

When LC5 is returned to its position No. 2 all the other stages will be in the second position, which was the original position in the sequence of operations.

While the fifth stage is brought from position No. 1 to position No. 2, which

70

only takes place at the particular moment preceding the restoration of the general original conditions, a positive pulse is generated by the left-hand plate of this fifth stage. This latter positive pulse is applied, as shown, through the small condensers JL and LL to the left-hand grid of HL. The rectifier-resistance network KL is added to shunt to the earth the undesirable negative pulse produced when the fifth stage is changed over from position No. 1 to position No. 2. The result is that the positive pulse on the left hand plate due to the change-over of this fifth stage from position No. 1 to position No. 2, restores HL to its "off" position which, in turn, immediately blocks the oscillator DL from the input of the frequency divider. The counting operation then ceases and the original conditions are restored throughout the circuit; these conditions remain until the arrival of the next combination of pulses from the transmitter, at which moment a similar cycle of operations is repeated.

As already stated, the variations in the duration of the pulses generated by HL are the inverse substantially of the variations of the amplitudes which they represent of the input speech wave at BT, Fig. 4. By means of the coupling condenser TL these pulses are transferred to the plate current of the valve UL. The current variations in the output of UL are inverted by the transformer between UL and VL and the mean current in the output of this transformer, obtained by eliminating the pulse frequencies by the low-pass filter VL, will, consequently give a substantially exact reproduction of the original speech at the transmitter, which speech will thus have a background noise of 70 db. or less since it is due to the noise in one stage only, the remaining valves acting as simple relays, wholly unaffected by the circuit noise.

The invention has been described in the case of particular embodiments, but it is clear that it is in no way limited thereto, but on the contrary is capable of numerous adaptations and modifications without departing from the scope thereof.

Having now particularly described and ascertained the nature of our said invention and in what manner the same is to be performed, we declare that what we claim is:—

1. An electrical signalling system for transmitting complex wave forms, e.g. speech, wherein the wave form is scanned at the transmitter at predetermined instants and at these instants signal elements dependent on the wave form are transmitted to the receiver characterised

in this, that the amplitude range of the wave form to be transmitted is divided into a finite number of predetermined amplitude values according to the degree of fidelity required, the instantaneous amplitude value of the wave form to be transmitted at each predetermined instant being transmitted in a signal code representing the nearest predetermined amplitude value, above or below said instantaneous amplitude value.

2. A signalling system for transmitting complex wave forms according to claim 1, characterised in this, that the number of elements in said code is the same as the number of predetermined amplitude values.

3. A signalling system for transmitting complex wave forms according to claim 1, characterised in this, that the elements of said code consist of a plurality of separate signal elements, the number of combinations of the code signal elements being equal to or greater than the finite number required for reproducing the complex wave form within the desired limits of accuracy.

4. A signalling system for transmitting complex wave forms according to claim 2 or 3, characterised in this, that the signal elements or the combinations of the signal elements are automatically converted at the receiver into variable duration pulses which are used to produce approximately depending upon the degree of fidelity required in the reproduced signal, the instantaneous amplitude at the instant in question of the complex wave form.

5. A signalling system for transmitting a complex wave form according to claim 3, characterised in this, that the plurality of separate signal elements are transmitted over "m" separate transmitting channels each channel consisting of a carrier wave of different frequency from the others.

6. A signalling system for transmitting a complex wave form according to claim 3, characterised in this, that the signal elements consist of a plurality (n) of signals each having a different characteristic feature transmitted on a plurality (m) of separate channels.

7. A signalling system for transmitting a complex wave form according to claim 6, characterised in this, that the plurality of different signals transmitted on each channel comprise like signals of different predetermined amplitudes.

8. A signalling system for transmitting a complex wave form according to claim 3, characterised in this, that the plurality of separate signal elements consist of pulses transmitted on "m" 130

separate channels each utilising a carrier wave of the same frequency, said channels being brought successively and cyclically into use by means of distributor arrangements.

5 9. A signalling system for transmitting a complex wave form according to claims 6 and 7, characterised in this, that the number of signal elements
10 transmitted in each channel is so chosen that at the receiver a predetermined peak value of circuit noise is incapable of effecting the occurrence or non-occurrence of the series of operations associated with
15 the particular signal code transmitted at that instant.

10. A signalling system for transmitting a complex wave form according to claim 6, characterised in this, that the
20 signal elements in each channel consist of pulses of predetermined finite duration or pairs of pulses of short fixed duration the pulses of a pair occurring at predetermined fixed periods of time.

25 11. A signalling system for transmitting a complex wave form according to claims 5 and 6, characterised in this, that the signal elements in each channel consist of different predetermined phases of
30 the transmitted frequency.

12. A signalling system for transmitting a complex wave form according to claim 6, characterised in this, that the
35 signal elements in each channel consist of different predetermined frequencies.

13. A signalling system for transmitting a complex wave form according to claim 2, characterised in this, that the
40 arrangements at the sending end comprise a multivibrator circuit arrangement adapted to oscillate at a frequency outside the signal frequency range, means for applying the signal waves of complex
45 wave form in push-pull to the control grids of the multivibrator valve elements, and an oscillator loosely coupled to said multivibrator and arranged to oscillate at a frequency which is a multiple of the
50 multivibrator frequency, whereby the multivibrator can only begin an oscillation at the moment of reception of a peak voltage from said oscillator, the outputs from the two anode circuits of the multivibrator being passed through a resistance
55 shunted by a rectifier device, and the code signal elements being taken from across the terminals of said resistance.

14. A signalling system for transmitting a complex wave form according to
60 claim 2 in which the signal elements consist of pulses of different durations which are determined by the moment of inception of the pulse in the output circuit of a multivibrator at the transmitter and
65 controlled jointly by the input speech

wave and an auxiliary oscillator arranged to oscillate at a frequency which is a multiple for example, of the order of 100, of the multivibrator frequency, characterised in this, that the receiving
70 arrangements comprise a multivibrator arrangement to which the received pulse signals of varying duration are applied in parallel to the valve elements, thereof, and an auxiliary oscillator loosely
75 coupled to said valve elements, the said auxiliary oscillator being arranged to generate a frequency, equal to the frequency of the auxiliary oscillator at the transmitter, the rela-
80 tive phase relationship between the auxiliary oscillators at the transmitter and receiver being such that a pulse transmitted at the moment of a peak of the auxiliary oscillation of the trans-
85 mitter arrives at the receiver also at a peak of the auxiliary oscillation of the receiver the output from said receiver multivibrator being passed through a low pass filter.

15. A signalling system for transmitting complex wave forms according to claim 6, in which the signal code comprises two signal elements ("on" and
95 "off") in each of "m" channels, characterised in this, that the transmitting arrangements comprise arrangements for modulating the duration of rectangular current pulses according to
100 the amplitude of the complex wave at predetermined instants, a frequency dividing arrangement used as a cycle counter and comprising multivibrator stages connected in cascade, one for each
105 channel, arrangements for rendering said multivibrator stages selectively operated in response to particular durations of said rectangular pulses, arrangements for returning the multivibrator stages to their normal condition at the end of each
110 rectangular pulse and oscillation generators of different frequencies controlled by said multivibrators to feed to the transmitting medium pulses of different carrier frequencies in accordance with the multi-
115 vibrator stages operated.

16. A signalling system for transmitting complex wave forms according to claim 6, in which the signal code comprises two elements ("on" and "off")
120 over each of "m" channels of different carrier frequencies and in which at the transmitter pulses of one or more of the different carrier frequencies are generated simultaneously, the combination of said
125 pulses at successive instants depending upon the amplitude of the input complex wave at said predetermined instants, characterised in this, that the receiver comprises a frequency dividing arrange-
130

ment used as a cycle counter and having
"m" multivibrator stages connected in
cascade, arrangements for rendering said
dividing arrangement operative on recep-
5 tion of said pulses from the transmitter,
means for changing over respective ones
of the multivibrator stages from one
stable equilibrium position to the other
under the control of the received pulses
10 of different carrier frequencies so as to
vary the durations of the cycle counting
operation according to the combinations
of the multivibrators so operated, said
frequency dividing arrangement being
15 adapted to control a device for producing
rectangular pulses of respective durations
corresponding to the respective combina-

tions of pulses of different carrier fre-
quencies received and a low pass filter to
which successive pulses from the said 20
device are applied.

17. A signalling system for transmit-
ing complex wave forms according to
claim 1, substantially as hereinbefore
described with reference to Figs. 2 and 25
3, or 4 and 5, of the accompanying
drawings.

Dated this 23rd day of October,
A.D. 1939.

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Fig. 1.

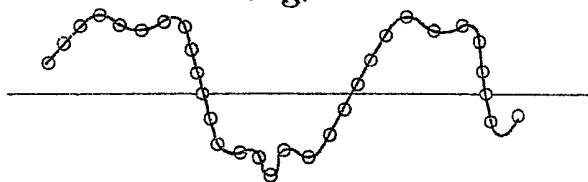


Fig. 2.

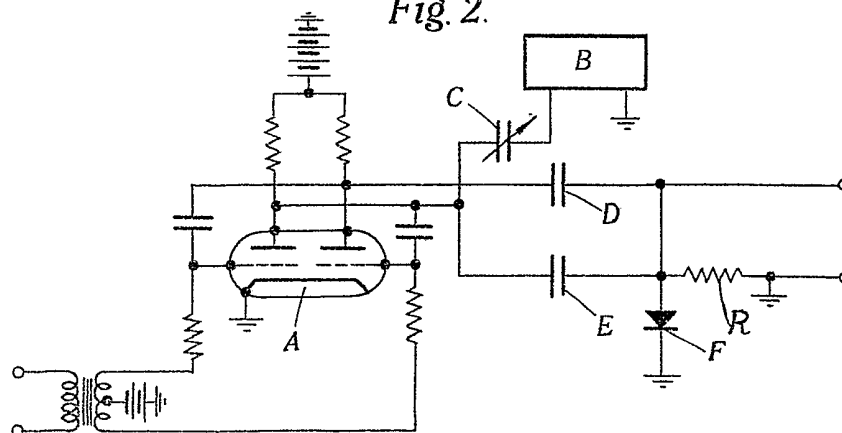
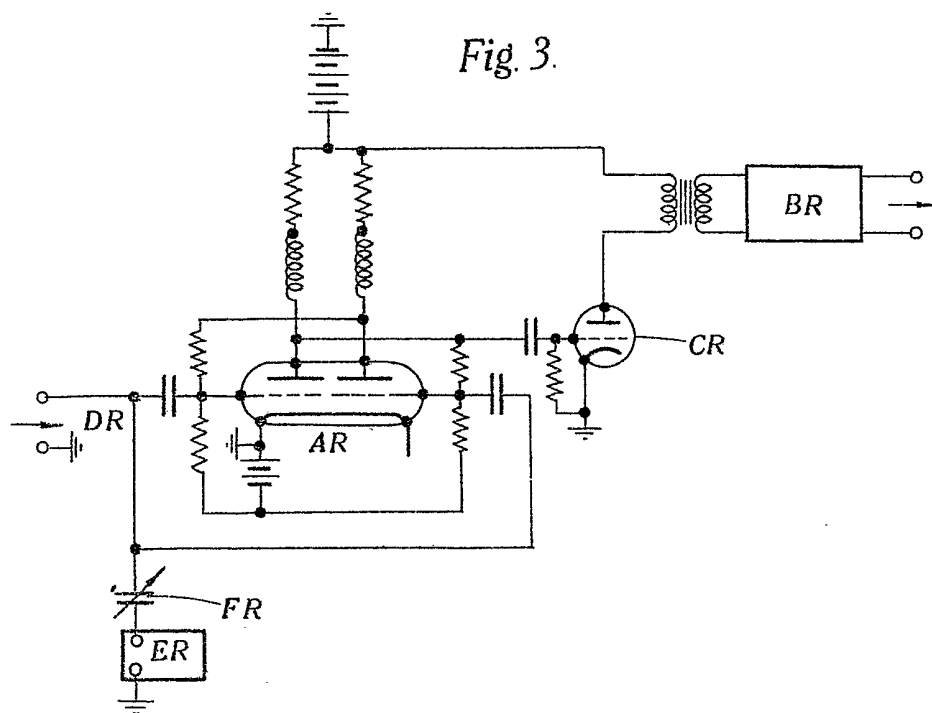


Fig. 3.



[This Drawing is a reproduction of the Original on a reduced scale.]

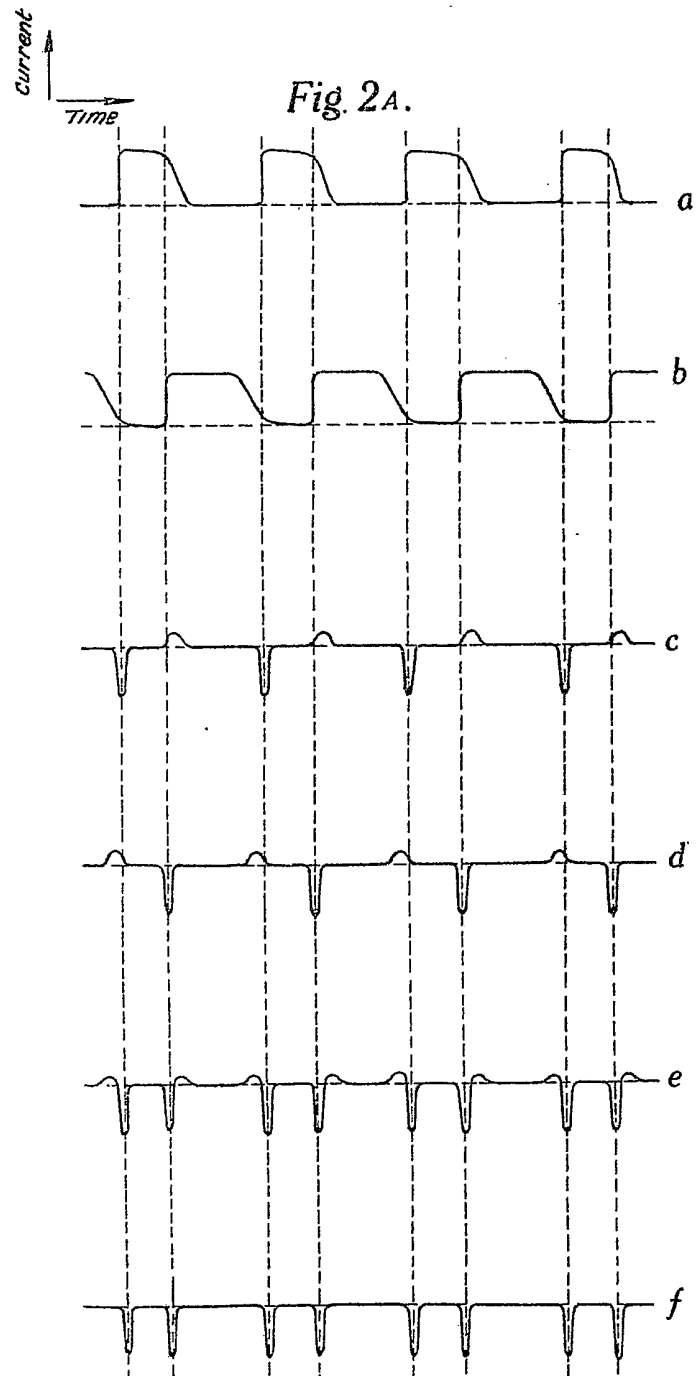


Fig. 1.

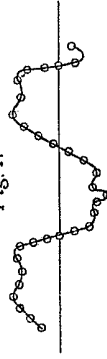


Fig. 2.

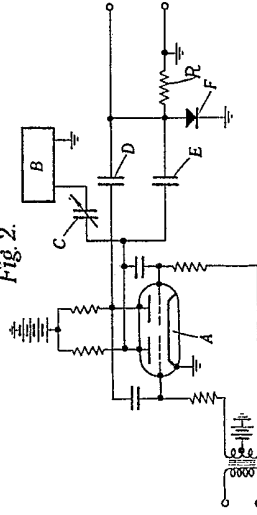


Fig. 3.

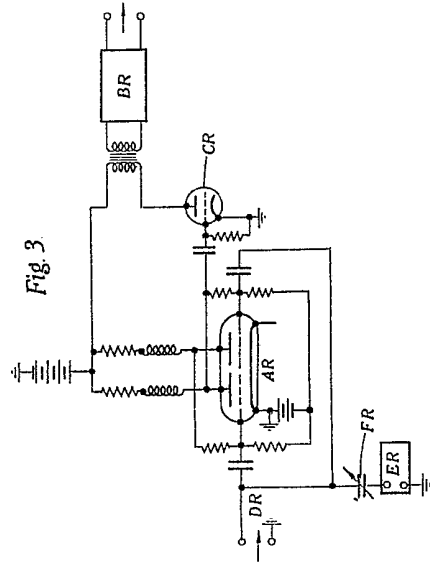
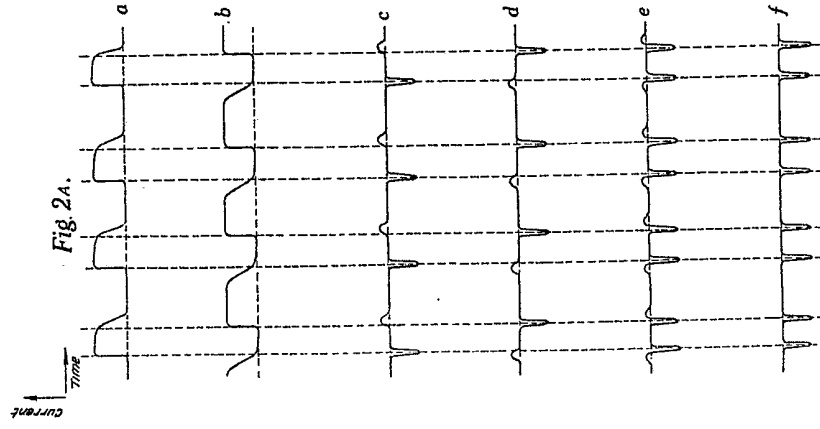


Fig. 2A.



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Fig. 4.

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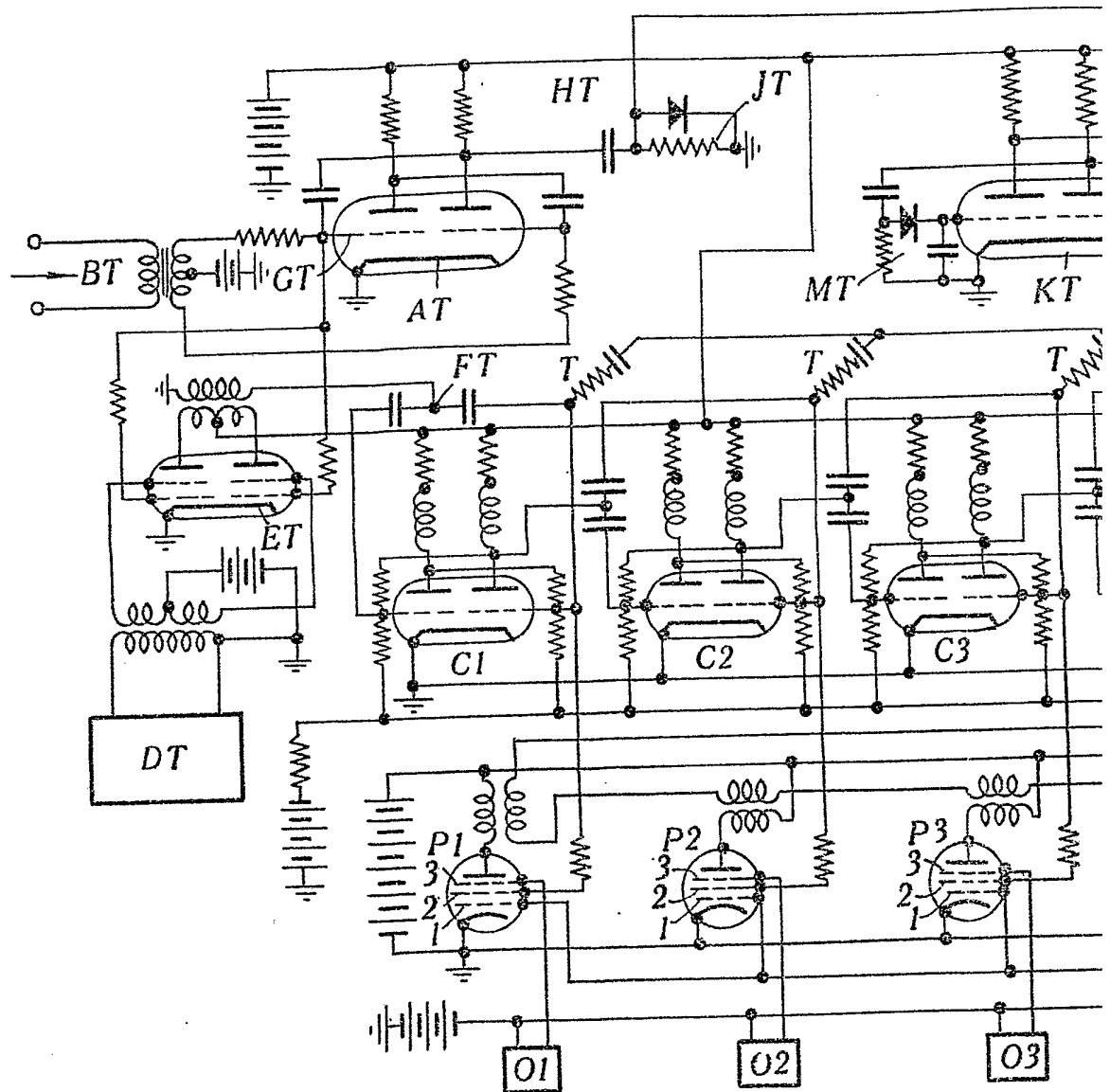


Fig. 4.

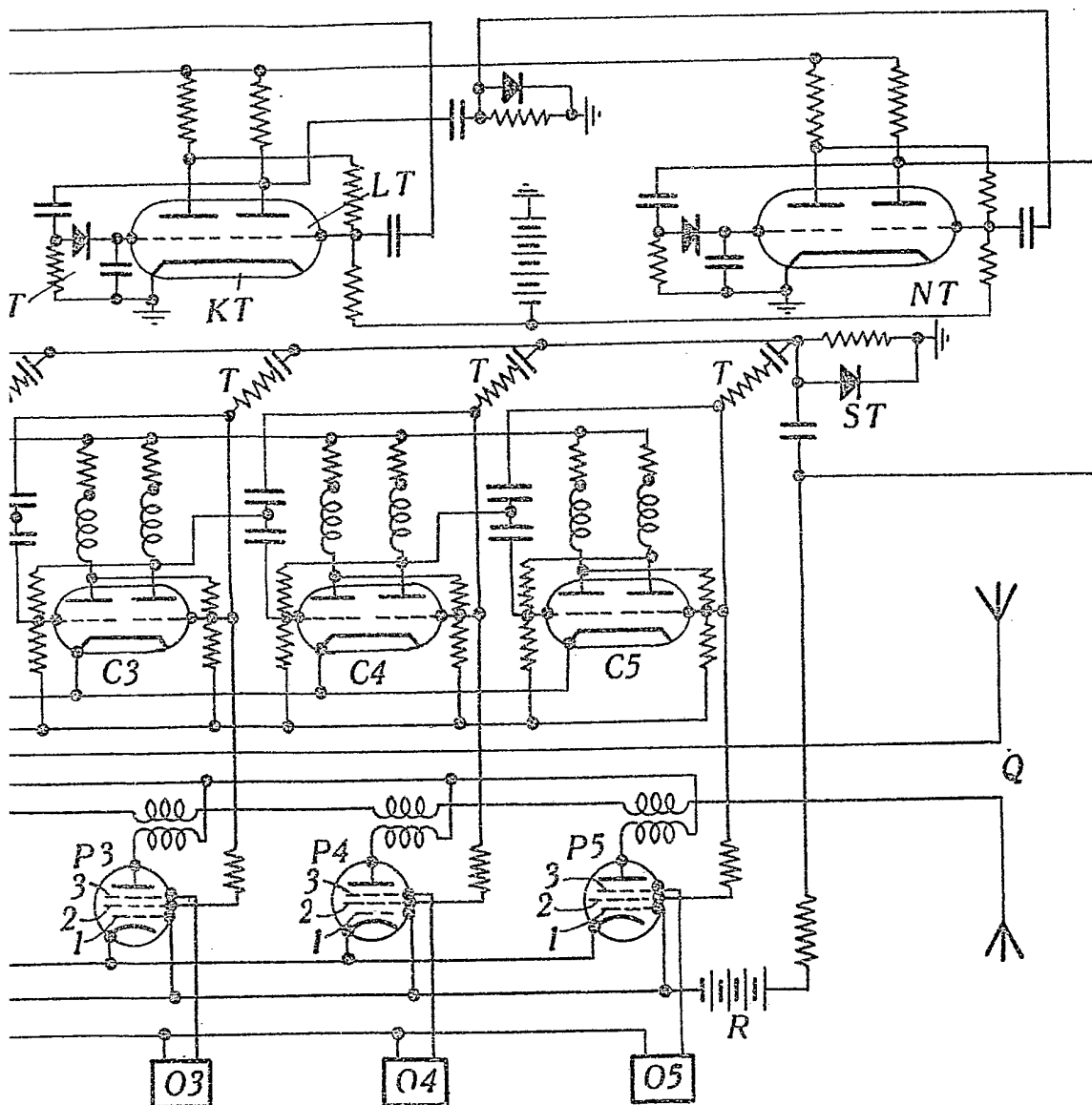
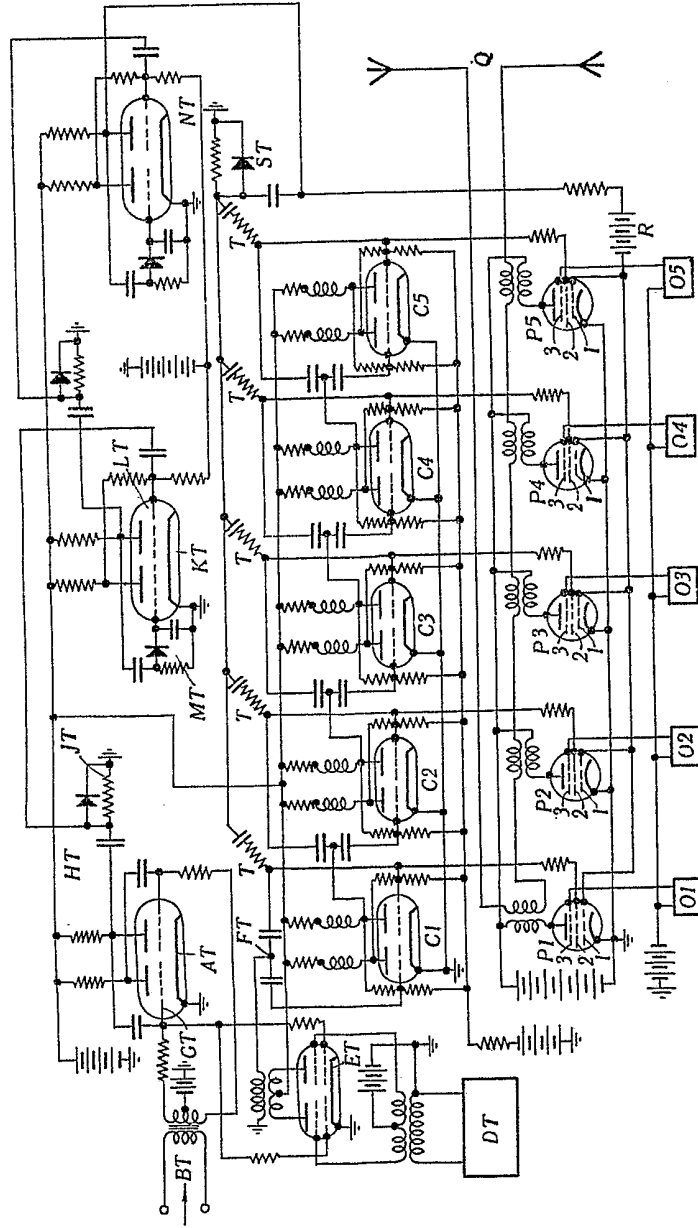
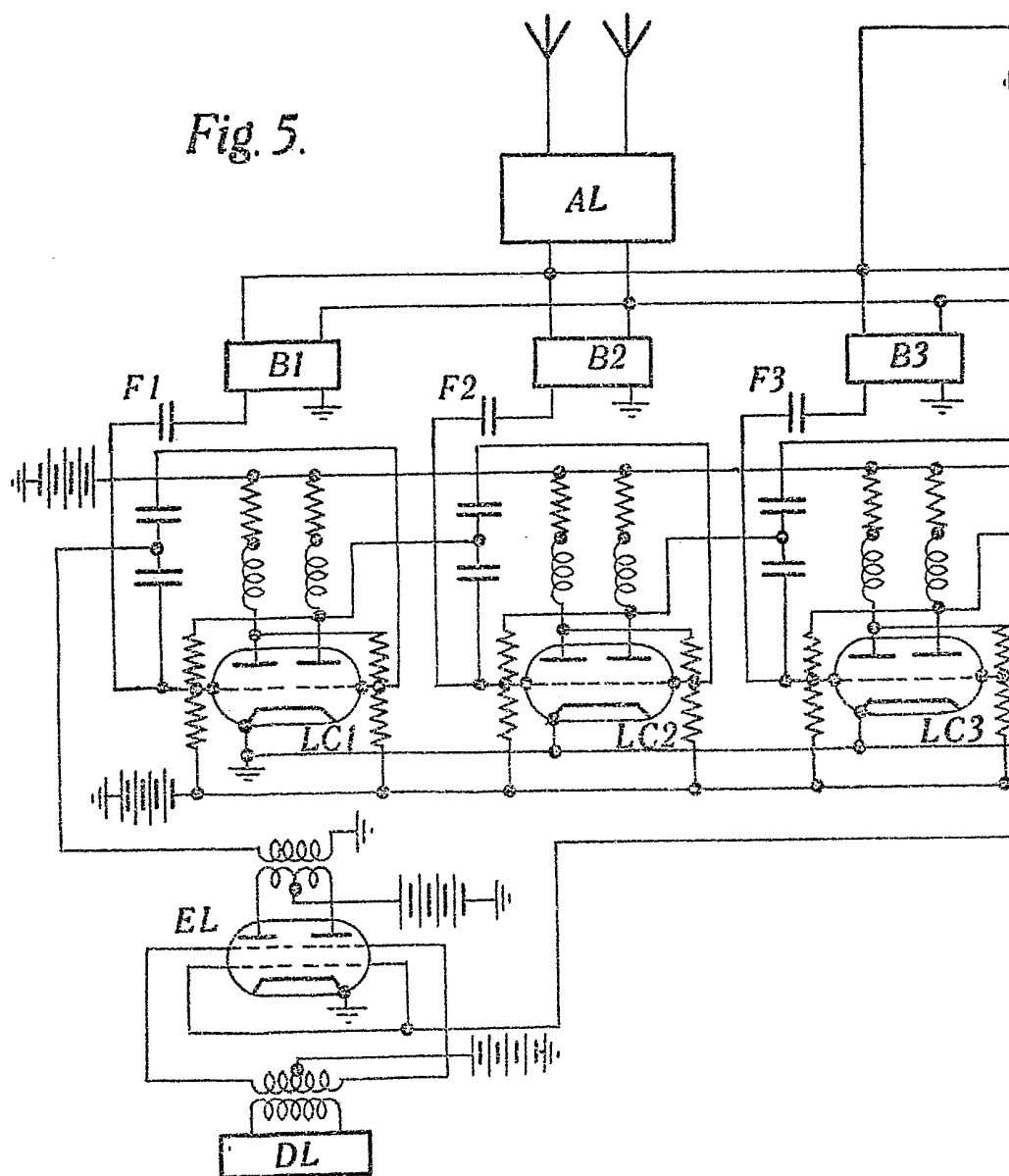


Fig. 4.

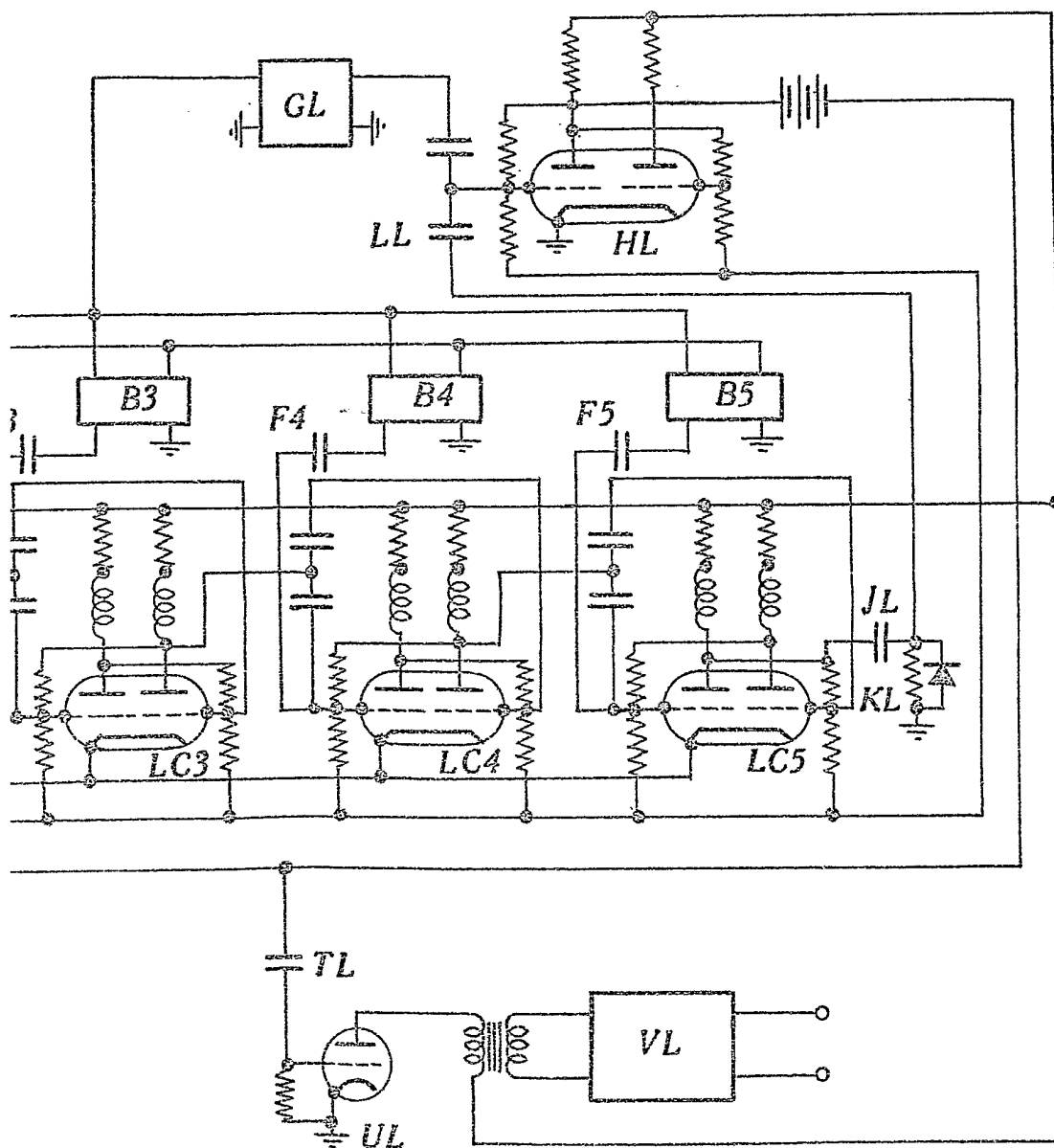


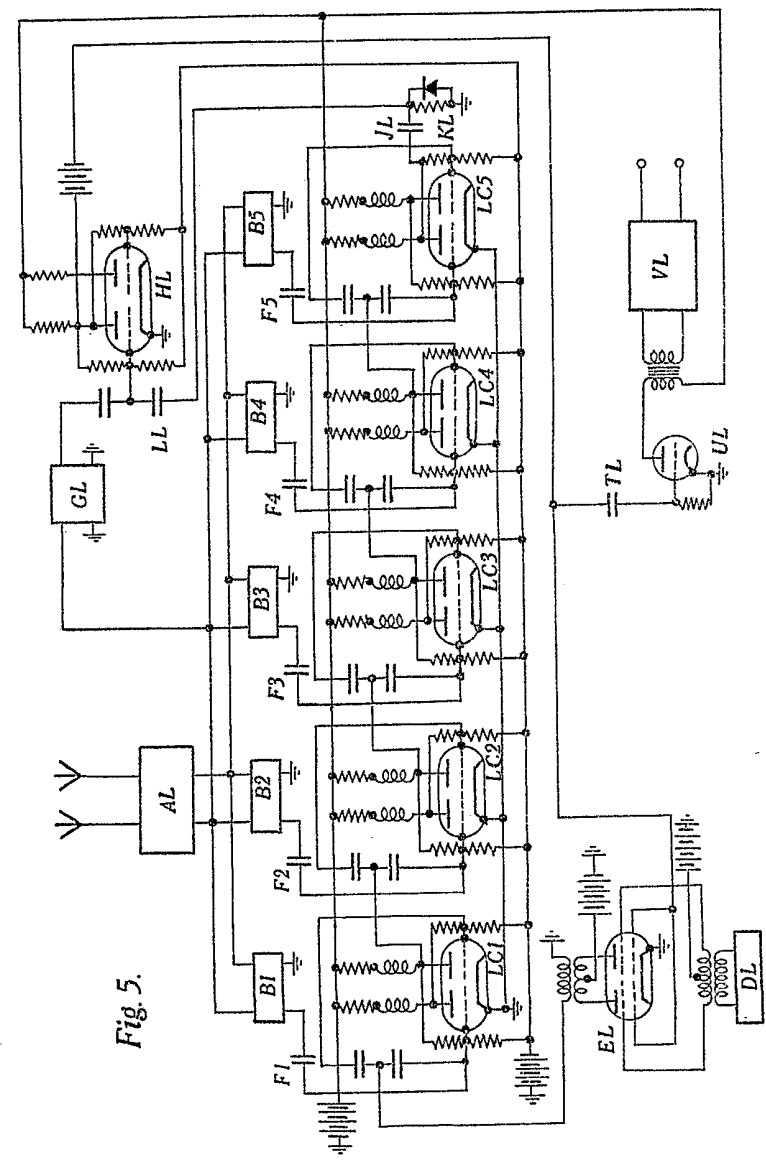
[This Drawing is a full-size reproduction of the Original.]

Fig. 5.



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